

SPARC

Single-pass scaling for motion forecasting with Bayesian last layers

Sakif Hossain, Julian Teusch, Jörg P. Müller
Sakif Hossain*, Julian Teusch*, Jörg P. Müller

Equally contributed
European Conference on Computer Vision (ECCV) 2026, Malmö,
Sweden

CVPR 2026 Spotlight
European Conference on Computer Vision (ECCV) 2026

Malmö, Sweden, September 8–12, 2026

Presentation date: *[insert date]*

Motion Forecasting Needs More Than Points

- Input: a short observed 3D pose sequence [16, 30].
- Output: a plausible future trajectory.
- Safety-critical HRI needs predictive envelopes, not only means [21, 23, 37].
- Target: uncertainty that is efficient, calibrated, and structured.

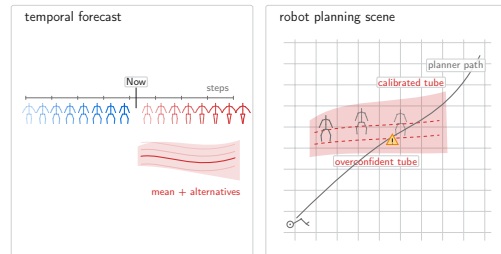


Fig. 1: Observed motion, forecast uncertainty, and a robot path motivate calibrated tubes.

PROBLEM

Three Requirements Rarely Meet

- ▶ **Epistemic:** detect weak training support [20].
- ▶ **Aleatoric:** represent genuine motion ambiguity [20].
- ▶ **Calibrated:** reach target coverage, e.g., 95% [1, 15].

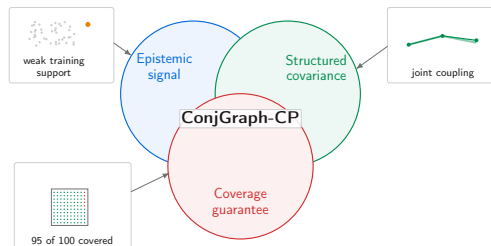


Fig. 2: The method combines epistemic signal, structured covariance, and conformal coverage.

Existing Methods Face a Deployment Trade-off

- ▶ MC dropout and ensembles often require repeated stochastic passes [13, 22].
- ▶ Generative forecasters use samples, hypotheses, or diffusion loops [2, 38, 47].
- ▶ Calibration-only methods provide coverage but no epistemic risk signal [1, 35].
- ▶ Factorized Gaussian heads lose temporal and skeletal dependence [3, 10, 36].

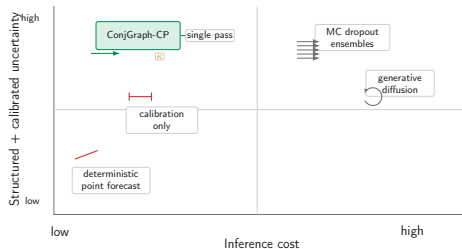


Fig. 3: Deployment methods trade inference cost against structured calibrated uncertainty.

SPARC as an Uncertainty Layer

1. Deterministic siMLPe/DCT backbone $\rightarrow$ mean forecast [16].
2. Bayesian last layer $\rightarrow \kappa_t(x)$ [11, 27].
3. Graph-temporal covariance $\rightarrow$ structured tubes [3, 10, 36].
4. Split conformal calibration $\rightarrow$ 95% marginal coverage [1, 26, 43].

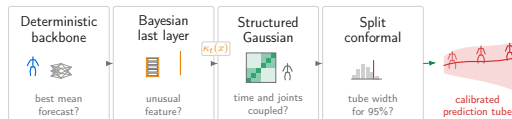
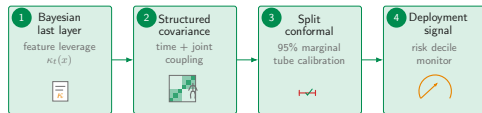


Fig. 4: SPARC wraps a deterministic forecaster with Bayesian scaling and split conformal calibration.

Paper Contributions

- **Structured Bayesian scaling:** $\kappa_t(x)\Sigma_{\text{str},t}(x)$ preserves covariance structure.
- **Single-pass uncertainty:** no MC sampling and no ensembles.
- **Cross-dataset gains:** best mean NLL rank and best combined MPJPE+NLL rank.
- **Risk monitoring:** on H36M, the top- κ decile has $1.79\times$ average MPJPE.



single-pass calibrated uncertainty: no MC dropout, no ensemble sampling

Fig. 5: The contribution chain turns feature leverage into single-pass calibrated uncertainty.

Task and Notation

- Observed prefix: $x_{1:T} \in \mathbb{R}^{T \times C}$.
- Future trajectory: $y_{1:H} \in \mathbb{R}^{H \times C}$.
- $C = 3J$ coordinates for J joints.
- Training uses displacements relative to the last observed pose x_T .

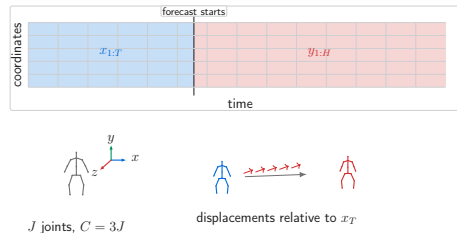


Fig. 6: Observed prefixes and future targets are represented as time-by-joint trajectory blocks.

Desired Output: Prediction Tubes

$$\hat{C}_\alpha(x) = \prod_{t,j,d} [\hat{\mu}_{t,j,d}(x) \pm q_{t,j} \hat{\sigma}_{t,j,d}^{\text{CP}}(x)] .$$

- ▶ $q_{t,j}$ is a split-conformal factor [26, 43].
- ▶ The paper uses $\alpha = 0.05$ for 95% coverage.
- ▶ The guarantee is marginal, not automatically trajectory-level [12, 25].

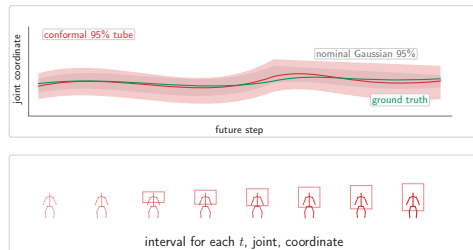
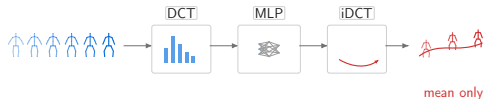


Fig. 7: Prediction tubes wrap the mean forecast with conformal marginal intervals.

Backbone: siMLPe/DCT

- ▶ The time axis is represented by DCT coefficients [16].
- ▶ An MLP processes coefficient-domain features.
- ▶ iDCT maps outputs back to the time domain.
- ▶ MeanFT provides a strong deterministic forecast.



deterministic backbone: uncertainty is added after the mean forecast

Fig. 8: The siMLPe-style DCT backbone maps pose histories to a deterministic future mean.

Time-Domain Design Vector

$$\hat{\mu}_t(x) = B_t^\top \phi_t(x), \quad \phi_t(x) = \begin{bmatrix} g_t(x) \\ s_t \end{bmatrix}.$$

$$g_t(x) = \sum_{n=1}^N A_{t,n} h_n(x), \quad s_t = \sum_{n=1}^N A_{t,n}.$$

- One design vector per forecast horizon step.
- The leverage of $\phi_t(x)$ later determines $\kappa_t(x)$.

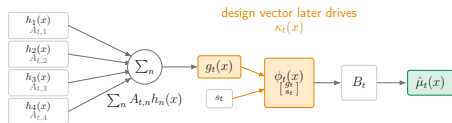
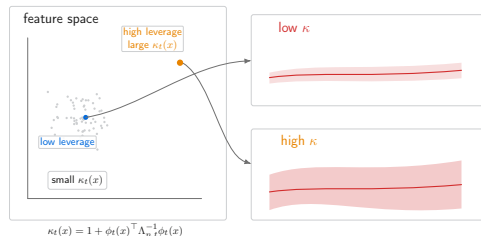


Fig. 9: DCT-domain features are aggregated into time-specific design vectors.

Epistemic Uncertainty as Feature Leverage

$$\kappa_t(x) = 1 + \phi_t(x)^\top \Lambda_{n,t}^{-1} \phi_t(x).$$

- ▶ Low leverage: test features lie in a dense training region.
- ▶ High leverage: weak feature-space training support.
- ▶ This is representation-dependent, not a universal OOD detector [33].



Bayesian last-layer background: [11, 27, 34, 44].

Fig. 10: Feature leverage separates dense training support from higher epistemic risk regions.

Proposition 1: Core Formula

Model:

$$y_t = B_t^\top \phi_t(x) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_{\text{str},t}(x)).$$

Matrix-normal prior [10]:

$$B_t \mid \Sigma_{\text{str},t} \sim \text{MN}(B_{0,t}, \Lambda_0^{-1}, \Sigma_{\text{str},t}).$$

Result:

$$y_t \mid x, D \sim \mathcal{N}(\mu_{\text{post},t}(x), \kappa_t(x) \Sigma_{\text{str},t}(x)).$$



Fig. 11: Bayesian last-layer scaling enlarges covariance while preserving its structure.

Scale Epistemics, Preserve Structure

Weak alternative

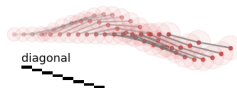
- Diagonal or isotropic uncertainty treats coordinates as largely independent.

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$$\Sigma_{\text{hyb},t}(x) = \kappa_t(x) \Sigma_{\text{str},t}(x).$$

- Scale increases; correlations are preserved.

factorized uncertainty



structured uncertainty

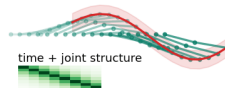


Fig. 12: Structured scaling preserves coherent skeletal uncertainty compared with diagonal inflation.

Offline Fit and Single-Pass Inference

Offline

$$S_{xx,t} = \sum_i \phi_t(x_i) \phi_t(x_i)^\top.$$

$$\Lambda_{n,t}^{-1} = (S_{xx,t} + \lambda_0 I)^{-1}.$$

Inference

- ▶ one forward pass,
- ▶ a quadratic form for $\kappa_t(x)$,
- ▶ no MC sampling and no ensembles.

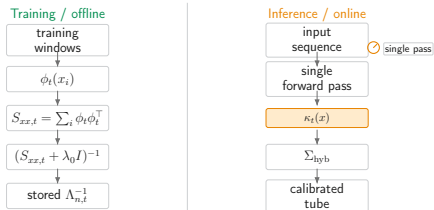


Fig. 13: Offline statistics enable one-pass online inference without sampling loops.

Stabilizing κ

$$\bar{\kappa}(x) = \frac{1}{H} \sum_{t=1}^H \left(1 + \phi_t(x)^\top \Lambda_{\text{glob}}^{-1} \phi_t(x) \right).$$

$$\tilde{\kappa}_t(x) = (1 - \rho) \kappa_t(x) + \rho \bar{\kappa}(x).$$

- Shrinkage smooths horizon-wise scales.
- Per-joint $\kappa_j(x)$ scales CP tubes, not the Gaussian NLL core.



Fig. 14: Shrinkage stabilizes horizon-wise leverage while keeping the Gaussian NLL core unchanged.

Structured Aleatoric Covariance

Matrix-normal residual model [10]

$$R(x) = y_{1:H} - \hat{\mu}_{1:H}(x),$$

$$R(x) \sim \text{MN}(0, \Sigma_T(x), \Sigma_C(x)).$$

Joint-graph precision [3, 36]

$$Q_J(x) = \tau(x)L_{\text{joint}} + \epsilon(x)I_J,$$

$$\Sigma_C(x) = (Q_J(x) \otimes I_3)^{-1}.$$

Σ_T captures time; Σ_C captures skeletal coupling.

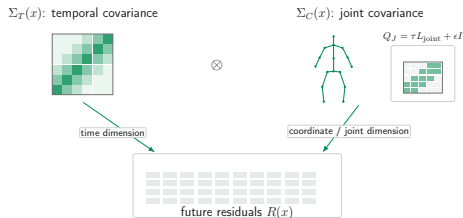


Fig. 15: MN-GraphJ combines temporal covariance with skeletal joint coupling.

Hybridization

$$D_{\kappa}(x) = \text{diag} \left(\sqrt{\tilde{\kappa}_t(x)} \right)_{t=1}^H.$$
$$\Sigma_T^{\text{hyb}}(x) = D_{\kappa}(x) \Sigma_T(x) D_{\kappa}(x)^{\top}.$$
$$\Sigma^{\text{hyb}}(x) = \Sigma_T^{\text{hyb}}(x) \otimes \Sigma_C(x).$$

- ▶ Correlations remain intact.
- ▶ Variance increases where $\tilde{\kappa}_t$ is high.

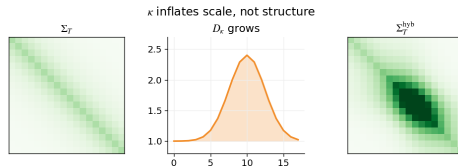


Fig. 16: Hybridization inflates temporal variance according to κ while retaining correlations.

Split Conformal Calibration

Calibration scores [1, 26, 43]:

$$s_{i,t,j,d} = \frac{|y_{i,t,j,d} - \hat{\mu}_{t,j,d}(x_i)|}{\hat{\sigma}_{t,j,d}^{\text{CP}}(x_i)}.$$

Quantile:

$$q_{t,j} = Q_{1-\alpha}(\{s_{i,t,j,d}\}_{i,d}).$$

Tube:

$$\hat{\mu}_{t,j} \pm q_{t,j} \hat{\sigma}_{t,j}.$$

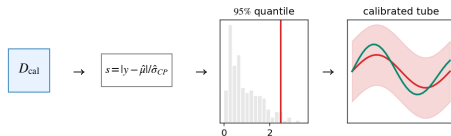


Fig. 17: Split conformal calibration converts residual scores into 95 percent prediction tubes.

What the Guarantee Says – and Does Not Say

Guaranteed under exchangeability

[26, 43]

$$\Pr\{y_{t,j} \in \hat{C}_{t,j}(X)\} \geq 1 - \alpha.$$

Not guaranteed

- ▶ strong conditional coverage [14],
- ▶ shift-robust guarantees [33],
- ▶ automatic trajectory-level coverage.

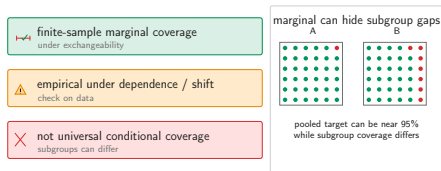


Fig. 18: Marginal validity can hold even when subgroup coverage differs.

SPARC in Nine Steps

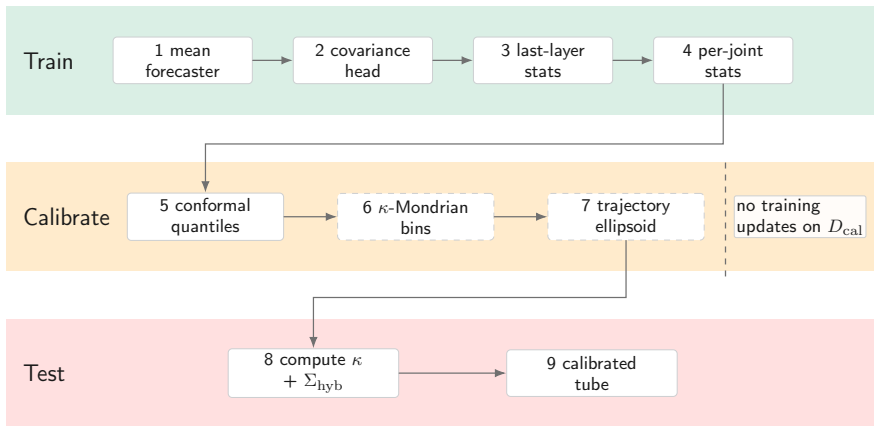


Fig. 19: The algorithm separates training, calibration, and deployment without updating on calibration data.

Optional Extensions

- ▶ κ -Mondrian CP: separate quantiles for risk bins [4, 42].
- ▶ Trajectory CP: structured Mahalanobis or whitened-residual score.
- ▶ Goal: risk-adaptive or more coherent set-valued prediction [24, 32, 48].

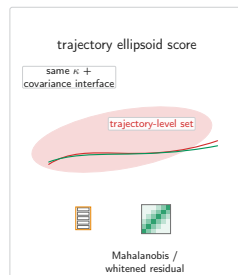
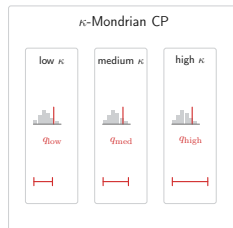


Fig. 20: Optional extensions reuse the same interface for Mondrian bins and trajectory-level scores.

Datasets

- ▶ Primary benchmark: Human3.6M [19].
- ▶ Additional motion/MoCap datasets: AMASS, LaFAN1, CMU-MoCap, 3DPW [5, 17, 28, 41].
- ▶ HRI-style datasets: CHICO, HA4M, AnDy [8, 31, 37].
- ▶ Non-H36M protocol: $T = 50$, $H = 25$.

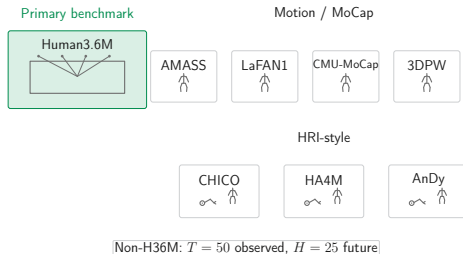


Fig. 21: The evaluation spans controlled MoCap, in-the-wild motion, and HRI-style datasets.

Metrics

- ▶ MPJPE: mean per-joint position error.
- ▶ FDE: final-horizon MPJPE.
- ▶ NLL: per-element Gaussian negative log-likelihood.
- ▶ Cov95/W95: nominal coverage and interval width.
- ▶ Cov95(CP)/W95(CP): conformalized coverage and width.

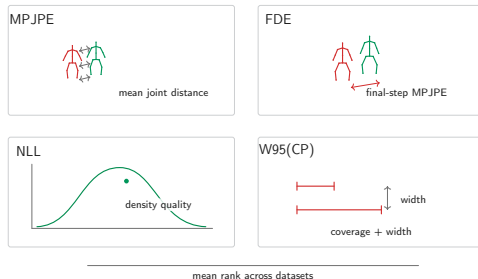


Fig. 22: Metrics compare point accuracy, density quality, empirical coverage, and tube width.

TABLE 1

Main Result: Mean Ranks

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- ▶ $MR_{NLL} = 1.00$.
- ▶ $MR_{MPJPE+NLL} = 2.69$.
- ▶ $MR_{W95(CP)} = 2.50$.
- ▶ $MR_{MPJPE} = 4.39$.

Model	MPJPE	NLL	W95(CP)	MPJPE+NLL
MeanFT	2.22	6.11	–	4.17
Bridge-CQR	4.61	<u>3.00</u>	9.11	<u>3.81</u>
Deep ensemble	8.22	5.56	5.39	6.89
CoMusion	6.06	6.44	3.78	6.25
SPARD	7.11	6.56	<u>3.39</u>	6.83
SPARC	<u>4.39</u>	1.00	2.50	2.69

Table 1: Selected mean-rank rows; the next slide gives the full cross-dataset table.

Lower rank is better. SPARC is strongest on density and the combined point+density trade-off.

LOWER IS BETTER

Full Table 1: Cross-Dataset Mean Ranks

Model	MR MPJPE	MR NLL	MR W95(CP)	MR MPJPE+NLL
siMLPe (base) [16]	7.89	10.67	—	9.28
MeanFT (siMLPe fine-tune) [16]	2.22	6.11	—	4.17
AuxTasks (50/50) [45]	9.06	11.78	—	10.42
Symplectic (50/50) [6]	16.22	15.78	—	16.00
DeFeeNet (50/50) [39]	7.61	10.44	—	9.03
GSPS (50/50) [29]	7.67	9.00	5.44	8.33
HumanMAC (50/50) [7]	10.22	10.00	—	10.11
SeSGCN (teacher)	13.39	11.78	—	12.58
Deep ensemble [22]	8.22	5.56	5.39	6.89
Bridge-CQR [35]	4.61	<u>3.00</u>	9.11	<u>3.81</u>
SkeletonDiffusion (CVPR 2025) [9]	15.00	13.67	9.78	14.33
BeLFusion (ICCV 2023) [2]	9.56	8.00	5.89	8.78
TransFusion (RA-L 2024) [40]	12.44	9.78	4.78	11.11
CoMusion (ECCV 2024) [38]	6.06	6.44	3.78	6.25
SPARD (AAAI 2026) [49]	7.11	6.56	<u>3.39</u>	6.83
MotionMap (CVPR 2025) [18]	16.11	16.22	9.44	16.17
SLD-HMP (ECCV 2024) [46]	13.22	15.22	6.50	14.22
SPARC	<u>4.39</u>	1.00	2.50	2.69

Table 2: Mean ranks across dataset/protocol blocks; MPJPE+NLL averages the MPJPE and NLL ranks.

Coverage Must Not Come From Huge Tubes

- ▶ Best density quality in the benchmark.
- ▶ Efficient calibrated tubes: coverage at moderate width.
- ▶ Single forward pass.
- ▶ κ remains interpretable as an epistemic risk signal for deployment-oriented HRI monitoring [21, 23, 37].

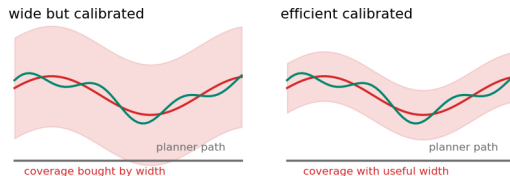


Fig. 23: Efficient calibration seeks target coverage without unnecessarily wide intervals.

Ablation: Component Contributions

Variant	κ_t	Hybrid	GraphJ	Coupled	MR MPJPE+NLL	MR W95(CP)
Base $\kappa(x)$	—	—	—	—	5.28	4.78
Timewise κ_t	✓	—	—	—	4.61	3.56
Hybrid MN	—	✓	—	—	3.94	4.00
Hybrid MN-GraphJ	—	✓	✓	—	2.47	4.78
Hybrid MN-GraphJ + timewise κ_t	✓	✓	✓	—	<u>2.53</u>	3.78
SPARC coupled κ_t	✓	✓	✓	✓	3.17	2.56

Table 3: Ablation over leverage, hybrid covariance, graph coupling, and coupled per-joint calibration.

FIGURE 3

Qualitative Uncertainty

- ▶ Gray: nominal 95% intervals, $\pm 1.960\sigma$.
- ▶ Red: split-conformal 95% intervals, $\pm q_{t,j}\sigma$.
- ▶ Green: ground truth.
- ▶ Higher mean κ creates visible epistemic inflation.

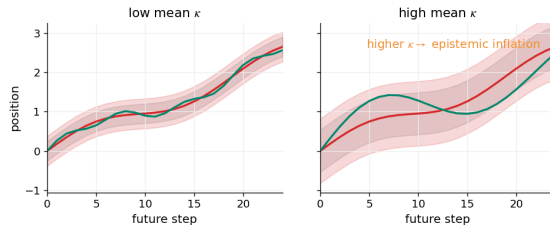


Fig. 24: Higher mean κ visibly expands calibrated intervals around the trajectory.

κ as a Risk Monitor

H36M diagnostic reported in the paper:

- ▶ Highest κ decile: $1.79\times$ average MPJPE.
- ▶ Filtering that decile reduces MPJPE by about 9%.
- ▶ The diagnostic supports κ as a deployment-time risk signal, not as a formal guarantee.
- ▶ Thresholds are tuned on held-out data and should be treated as policy parameters.

Why Single-Pass Matters

Inference requires:

- ▶ one backbone forward pass,
- ▶ horizon-wise quadratic forms for κ_t ,
- ▶ a structured Gaussian summary,
- ▶ conformal quantile lookup.

No MC dropout loop [13], deep ensemble loop [22], or diffusion sampling loop [2, 38].

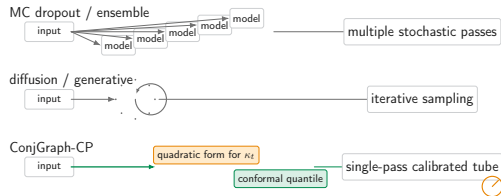


Fig. 25: Single-pass inference avoids repeated stochastic or generative sampling loops.

Takeaways and Limitations

Takeaways

- ▶ Analytic epistemic scaling via $\kappa_t(x)$.
- ▶ Structured trajectory covariance via $\Sigma_{\text{str},t}(x)$.
- ▶ Split conformal calibration gives 95% marginal tubes [26, 43].
- ▶ The method is strongest on density quality and the combined point+density trade-off.

Limitations

- ▶ κ is representation-dependent.
- ▶ Coverage is marginal, not strongly conditional [14].
- ▶ The graph covariance is a modeling choice.
- ▶ Online updates and coherent trajectory-level sets remain open.

Strict Split Separation

- ▶ Deterministic window permutation with seed 304.
- ▶ $n_{\text{cal}} = 512$ for split conformal calibration.
- ▶ $n_{\text{eval}} = 1024$ for final metrics.
- ▶ $\alpha = 0.05$ for 95% target coverage.
- ▶ Hyperparameters selected on held-out training windows.

seed 304

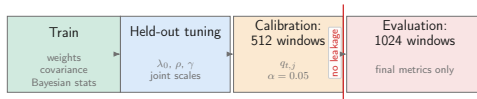


Fig. 26: Strict splits keep training, calibration, and evaluation windows separated.

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